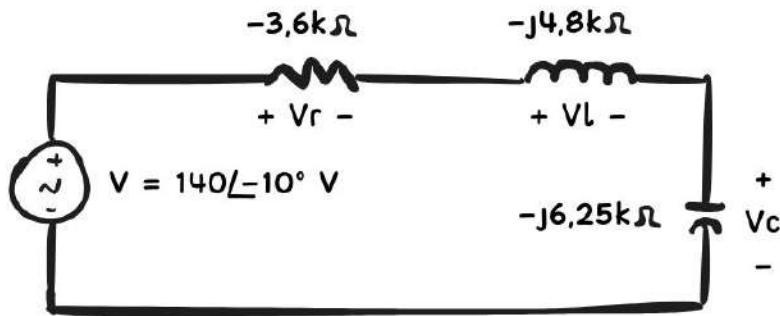


Questão 1)



Fonte de Tensão (V): $140\angle-10^\circ \text{ V}$

Resistor (R): $3,6\text{k}\Omega = 3600\angle 0^\circ \Omega$

Indutor (Xl): $j4,8\text{k}\Omega = 4800\angle 90^\circ \Omega$

Capacitor (Xc): $-j6,25\text{k}\Omega = 6250\angle -90^\circ \Omega$

Impedância Total (Z_t):

$$Z_t = R + jX_l + jX_c \quad \rightsquigarrow \quad Z_t = 3600 + j4800 + j6250$$

$$Z_t = 3600 - j1450 \Omega$$

$$|Z_t| = \sqrt{3600^2 + (-1450)^2} = 3881 \Omega$$

$$\theta = \arctan\left(\frac{-1450}{3600}\right) = -21.9^\circ$$

Corrente Total (I)

$$I = \frac{V}{Z_t} = \frac{140\angle-10^\circ}{3881\angle-21.9^\circ}$$

$$I = 0.0361\angle(-10^\circ - (-21.9^\circ)) \text{ A}$$

$$I = 36.1\angle 11.9^\circ \text{ mA}$$

Tensão no Resistor (Vr):

$$V_r = I \times R = (0.0361\angle 11.9^\circ) \times (3600\angle 0^\circ)$$

$$V_r = 129.96\angle 11.9^\circ \text{ V}$$

Tensão no Indutor (V_L):

$$V_L = I \times X_L = (0.0361 \angle 11.9^\circ) \times (4800 \angle 90^\circ)$$

$$V_L = 173.28 \angle 101.9^\circ \text{ V}$$

Tensão no Capacitor (V_C):

$$V_L = I \times X_L = (0.0361 \angle 11.9^\circ) \times (6250 \angle -90^\circ)$$

$$V_L = 225.6 \angle -78.1^\circ \text{ V}$$

Questão 2)

Impedância Total (Z_t):

$$V = 50 \angle 15^\circ \text{ V}$$

$$I = 10 \angle 70^\circ \text{ A}$$

$$Z_t = \frac{V}{I} = \frac{50 \angle 15^\circ}{10 \angle 70^\circ} = 5 \angle -55^\circ \Omega$$

$$Z_t = 5 \cos(-55^\circ) + j5 \sin(-55^\circ)$$

$$Z_t = 2.87 - j4.10 \Omega$$

$$\omega = 400 \text{ rad/s}, \quad L = 16 \text{ mH}$$

$$X_L = \omega L = 400 \times 0.016 = 6.4 \Omega \longrightarrow \text{Impedância do indutor é } +j6.4 \Omega$$

$$\text{Impedância Total}(Z_t) = Z_{\text{indutor}} + Z_{\text{outros}}$$

$$2.87 - j4.10 = j6.4 + Z_{\text{outros}}$$

$$Z_{\text{outros}} = 2.87 - j4.10 - j6.4$$

$$Z_{\text{outros}} = 2.87 - j10.5 \Omega$$

$$\downarrow$$

Resistor

$$2.87 \Omega$$

$$\downarrow$$

Capacitor

$$X_C = 10.5 \Omega$$

$$X_C = \frac{1}{\omega C} \longrightarrow 10.5 = \frac{1}{400 \times C}$$

$$C = \frac{1}{4200} \longrightarrow C = 2.38 \times 10^{-4} \text{ F}$$

Os outros elementos são: um resistor de 2.87Ω e um capacitor de $238 \mu\text{F}$.

Questão 3)

$$L = 1.5 \text{ H}$$

$$f = 60 \text{ Hz}$$

$$\omega = 2\pi f = 377 \text{ rad/s}$$

$$X_L = 2\pi fL = 2\pi \times 60 \times 1.5$$

$$X_L = 565.49 \Omega$$

$$X_C = X_L, \text{ então } X_C = 565.49 \Omega$$

$$X_C = \frac{1}{2\pi fC} \rightarrow C = \frac{1}{2\pi \times 60 \times 565.49} \rightarrow C = 4.69 \times 10^{-6} \text{ F}$$

Questão 4)

$$X_L = 2\pi fL = 2\pi \times 100 \times 0.2$$

$$V_{\text{rms}} = 120 \text{ V}$$

$$X_L = 40\pi = 125.66 \Omega$$

$$I_{\text{rms}} = 0.6 \text{ A}$$

$$L = 200 \text{ mH} = 0.2 \text{ H}$$

$$Z = R + jX_L \rightarrow X_L \text{ é cateto oposto e } |Z| \text{ é a hipotenusa}$$

$$f = 100 \text{ Hz}$$

$$\sin(\theta) = \frac{X_L}{|Z|} = \frac{125.66}{200} = 0.6283$$

$$|Z| = \frac{V}{I} = \frac{120}{0.6} = 200 \Omega$$

$$\theta = \arcsin(0.6283) = 38.9^\circ$$

$$\rightarrow Z = 200 \angle 38.9^\circ$$

Questão 5)

$$V = 240 \angle 0^\circ \text{ V}$$

$$I = 2 \text{ A} \text{ adiantada } 40^\circ \rightarrow 2 \angle 40^\circ \text{ A}$$

$$Z_I = 60 \angle 60^\circ \Omega$$

$$Z_{II} = ?$$

Impedância Total (Z_t):

$$Z_t = \frac{V}{I} = \frac{240 \angle 0^\circ}{2 \angle 40^\circ} = 120 \angle -40^\circ \Omega$$

$$Z_t = 120 \cos(-40^\circ) + j120 \sin(-40^\circ) = 91.93 - j77.13 \Omega$$

$$Z_I = 60 \cos(60^\circ) + j60 \sin(60^\circ) = 30 + j51.96 \Omega$$

$$Z_t = Z_I + Z_{II} \rightarrow Z_{II} = Z_t - Z_I$$

$$Z = 155.3 \angle -70.5^\circ$$

$$|Z_{II}| = 51.93^2 + (-146.41)^2 = 155.34 \Omega$$

$$\theta = \arctan\left(\frac{-146.41}{51.93}\right) = -70.47^\circ$$